Revealing Metabolic Interactions and Biotechnological Potential of Yeast–Microalgae Consortia Through Multi-Omics Integration

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Abstract

Yeast–microalgae consortia have mutually beneficial interactions, which include nutrient exchange, O_2–CO_2 exchange, and structural protection, providing biotechnological applications. This co-cultivation improves biomass yield, lipid accumulation for biofuels, and nutrient recycling in wastewater treatment, offering better performance compared to monocultures. However, large-scale implementation has several challenges, including strain compatibility, long-term stability, and industrial scalability. Integrative multi-omics research offers a holistic view of these interactions. Applying computational frameworks, including network-based integration and machine learning, would provide an opportunity to analyze metabolic exchanges, signaling pathways, and adaptive responses across omics layers. Particularly, deep learning uncovers nonlinear patterns and predicts and enhances our understanding of symbiotic dynamics, providing critical insights for optimizing microalgal co-cultivation. Thus, robust multi-omics integration in yeast–microalgae systems represent a promising approach to stress resilience, productivity, and metabolite yields, making the consortia a powerful platform for more efficient and potentially sustainable biotechnological applications.

Keywords: Yeast–microalgae consortia, Metabolic interactions, Multi-omics integration, Deep learning, Biotechnology applications

Introduction

There is an intricate cooperative relationship between yeast and microalgae within synthetic ecosystems. (1) Yeast produces vitamins and organic compounds that are beneficial for microalgal growth and survival, and in turn, microalgae make their environment nutrient-rich and physically protected. (1) Synthetic consortia have provided a number of advantages that result in biotechnological applications, which have increased interest in focusing on this research area.

In such consortia, microalgae provide oxygen and consume carbon substrates for the yeast, while the yeast releases carbon dioxide, which leads to increased algal photosynthesis. (2) These mutually beneficial interactions improve biomass production, enhance target product yields, and promote stress tolerance compared with monocultures. (2)Additionally, their nutrient and water requirements can be met using wastewater from municipal, agricultural, and industrial sources, which are cost-effective and sustainable in the long run. (2) This strategy is beneficial in both wastewater bioremediation and biofuel generation. It also enhances lipid accumulation and efficient nutrient recycling. (2) Despite the potential mentioned, there are several limitations in large-scale applications, including strain compatibility, stability of interactions over time, and scalability in industrial systems. (2)(3)(4) Thanks to integrative omics technologies, we have a more comprehensive view of biological systems compared to traditional approaches that use single-omics analyses, which provide limited insight. (5) Integrating multi-omics data using computational methods could be a powerful tool to uncover complicated molecular patterns and novel discoveries. (5)These computational approaches have primarily used disease mechanisms and cellular processes and provide a promising framework to examine metabolic interactions, signaling pathways, and adaptive responses in yeast–microalgae consortia. Applying such integrative analyses in these systems offers new opportunities to advance sustainable biotechnology applications. In doing so, systems-level inegration of omics data is critical.

Systems-Level Integration of Omics Data

Today, advanced technologies and available knowledge in omics data analysis and advanced computational methods allow us to study biological processes as a system, not with a fragmented view compared to traditional single omics methods.(5) Hence, Genomics, transcriptomics, proteomics, metabolomics, and data from other omics layers will be combined to help us better understand complicated biological pathways and processes.(5) (6) Computational multi-omics integration will enable us to produce meaningful discoveries such as cross-species metabolic pathways from these large and heterogeneous datasets. Network-based integrative approaches have a significant impact on multi-omics analysis. It offers a framework for demonstrating interactions within multiple different omics layers that could more accurately capture molecular complexity. (5)

Multi-Omics Integration Strategies

These analytical approaches can be classified into two major classes: multi-stage and multi-dimensional (multi-modal). In a multi-stage method, integration is performed in a stepwise manner. First, each omics layer is analyzed separately, and then statistical correlations between biological features from the datasets are investigated. This approach focuses on connections within and between omics layers and their relationships with the phenotype of interest. In contrast, multi-modal analytical methods integrate multiple omics profiles through simultaneous analysis. Some of these methods include machine learning–driven network-based approaches, network diffusion methods, and causality- or inference-based frameworks. (5)

Machine Learning in Multi-Omics Analysis

One of the most widely applied and effective tools in multi-omics studies is machine learning–driven network-based methods. These methods provide a powerful framework for multi-omics integration, enabling automatic model learning and accurate predictions. According to the study objective, supervised or unsupervised algorithms are selected for tasks such as prediction, feature discovery, and phenotype association mapping. (5)

Deep Learning for Complex Omics Data Patterns

With today's access to a vast amount of biological data, advanced informatics techniques are required. Most traditional statistical integrative analyses of large-scale biological data do not perform successfully in capturing any nonlinear patterns in multi-omics data analysis. Deep learning consists of many hidden layers and layer depth has a direct correlation with the capability of learning complex patterns and accuracy in predictions. Deep learning has the capability to encode and learn from heterogeneous and complex data in both supervised and unsupervised learning methods. Omics data analysis mainly has several problems, which deep learning performs adequately consisting of low signal, noise ratios and even datasets with a large number of variables and comparatively small numbers of samples or large analytical variance. (6) Despite classical graph-based ML assigning node and edge weights, in multi-view and deep learning, it is a self-teaching artificial intelligence analytical approach, applying cross-omics information to capture nonlinear patterns, which provide more opportunities to study complex systems. (6)(5)

Conclusions and Future Directions

Applying and further developing these powerful computational approaches for network-based integrative multi-omics analysis in yeast-microalgae co-cultivation systems would be critical to accessing the full potential of this research area. This consortia has complex metabolic interactions that regulate nutrient exchange, stress tolerance, and productivity. Exploiting integrative multi-omics methods, particularly by using machine learning and deep learning tools to explore and acquire better insight into cross-species metabolic connections that enhance lipid accumulation, biomass yield, or stress resilience, which are essential qualities in biotechnology applications. Although there are numerous advantages in multi-omics research, there are still some challenges, including heterogeneity, reproducibility, sparsity, and interpretation of results. This issue could be addressed by future research focusing on developing scalable and interpretable frameworks, hybrid models that can integrate paired and non-paired omics data, and standardized approaches that harmonize diverse datasets. Consequently, integration of deep learning architectures with network-based approaches is a particularly valuable method to obtain maximum accuracy in prediction while maintaining biological interpretability. Such advancements enhance the optimization of yeast-microalgae consortia as promising platforms for sustainable biotechnology.

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